



DiffusionGemma

Google's experimental open model for **4x faster** text generation using discrete text diffusion

- Open Source
- Apache 2.0
- 26B MoE

WHAT IS IT?

What is DiffusionGemma?

An **experimental open model** from Google that uses **text diffusion** - instead of predicting tokens one-by-one, it drafts and refines **entire blocks simultaneously**.

BUILT ON

- **Gemma 4 Architecture** open weights, runs locally
- **Gemini Diffusion Research** from Google I/O 2025

Like moving from a typewriter to a printing press - the entire page at once, not letter by letter.

DIFFUSION PROCESS · STEP 1 OF 3

Canvas of Noise

The model places **256 random tokens** on a canvas simultaneously. All positions start as noise - garbled, meaningless placeholders.

ZERP	CAL_	??N?	TOK_	IZ??	PIPE
FUNC	RET?	~N~	GENE	??E?	NULL
INIT	CALL	ERR_	SET_	~~~?	NONE
TK??	REM?	OUT_	???X	XFN?	FIN_

All 256 positions filled with random noise - generation begins



DIFFUSION PROCESS · STEP 2 OF 3

Iterative Denoising

Bidirectional attention runs across all 256 tokens in parallel. With each pass, more tokens resolve into real words - no waiting for prior tokens.

The	CAL_	mod	TOK_	eł	??N?
gen	FUNC	256	~ N~	tok	??E?
INIT	in	ERR_	par	~~~?	all
TK??	for	OUT_	ter	XFN?	FIN_

~50% of tokens resolved - all positions refining at the same time



DIFFUSION PROCESS · STEP 3 OF 3

Text Emerges

After multiple refinement passes, **all 256 tokens converge** to coherent, readable text. The full output is ready - faster than any sequential model.

The	mod	eL	gen	era	tes
256	tok	ens	in	par	all
eL,	si	mul	tan	eous	ly
for	fas	ter	out	put	✓

All 256 tokens resolved - complete output in one forward pass

SPEED

The Speed Numbers

1,107

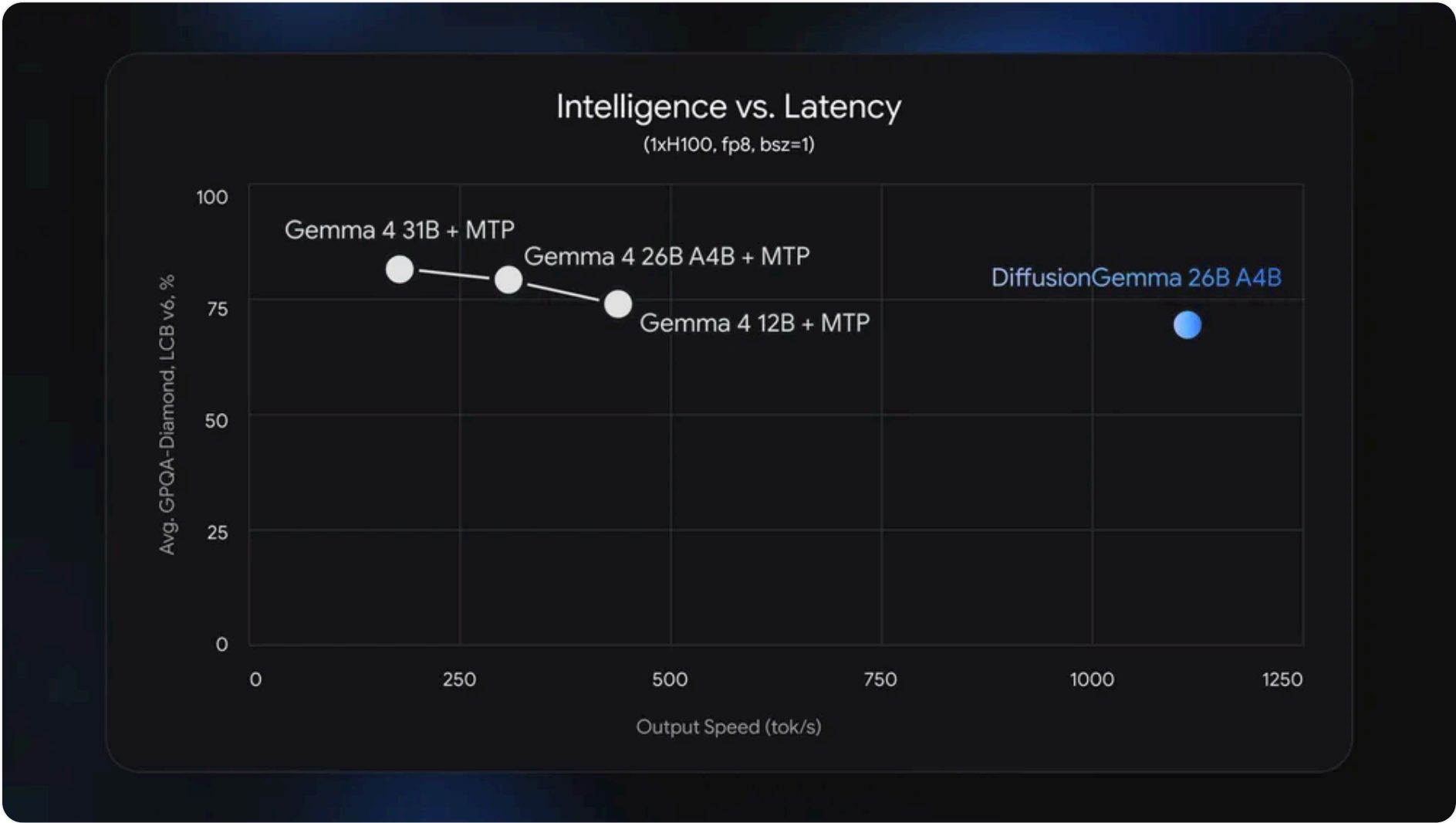
tok/s · H100

700+

tok/s · RTX 5090

4x

vs autoregressive

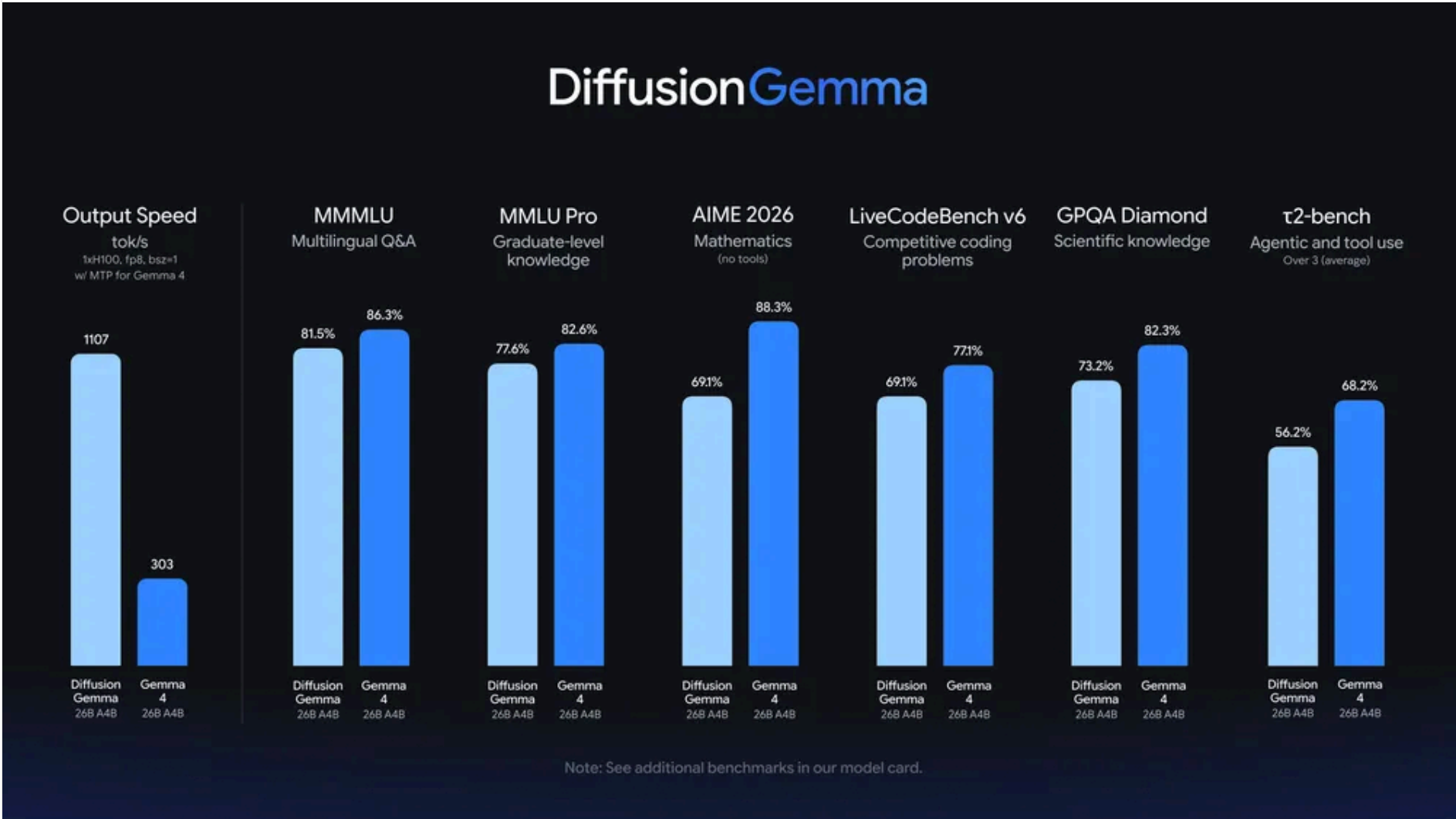


DiffusionGemma sits far right - same intelligence tier, dramatically higher output speed

BENCHMARKS

Speed Wins. Quality Trade-off.

DiffusionGemma outputs **1,107 tok/s vs 303 tok/s** for Gemma 4 - but trails slightly on reasoning benchmarks. Choose the right tool for the job.



FINE-TUNING · BEFORE

Same Starting Puzzle for Both

Both the base model and the finetuned model receive the identical puzzle with the same given clues (shown in gray). The difference is entirely in what happens next.

Finetuning DiffusionGemma to solve Sudoku

the same puzzle - given clues shown in gray

Base DiffusionGemma

.	5	.	6	4	.	.	.	.
.	8	4	.	.	.	.	.	2
.	.	3	.	.	9	4	.	1
.	6	.	.	3	.	9	8	.
3	.	8	.	.	.	.	1	5
7	9	1	.	6	5	2	4	3
5	1	6	2	.	4	.	3	.
.	.	9	.	.	.	7	.	4
.	.	.	.	.	.	.	.	6

starting puzzle

After finetuning

.	5	.	6	4	.	.	.	.
.	8	4	.	.	.	.	.	2
.	.	3	.	.	9	4	.	1
.	6	.	.	3	.	9	8	.
3	.	8	.	.	.	.	1	5
7	9	1	.	6	5	2	4	3
5	1	6	2	.	4	.	3	.
.	.	9	.	.	.	7	.	4
.	.	.	.	.	.	.	.	6

starting puzzle

analyticsvidhya.com

FINE-TUNING · AFTER

31 Cells Wrong vs. Solved

At denoising step 30/30: the base model gets **31 cells wrong**. The finetuned model solves it **perfectly** - green = correct, red = wrong.

Finetuning DiffusionGemma to solve Sudoku

denoising step 30 / 30 (green = correct cell, red = wrong cell)

Base DiffusionGemma

9	5	2	6	4	1	3	7	8
1	8	4	5	7	3	6	9	2
6	7	3	8	2	9	4	5	1
4	6	5	1	3	2	9	8	7
3	2	8	4	9	7	6	1	5
7	9	1	8	6	5	2	4	3
5	1	6	2	9	4	8	3	7
2	3	9	1	5	8	7	6	4
8	4	7	3	1	6	5	2	6

31 cells wrong

After finetuning

1	5	7	6	4	2	3	9	8
9	8	4	7	1	3	5	6	2
6	2	3	5	8	9	4	7	1
2	6	5	4	3	1	9	8	7
3	4	8	9	2	7	6	1	5
7	9	1	8	6	5	2	4	3
5	1	6	2	7	4	8	3	9
8	3	9	1	5	6	7	2	4
4	7	2	3	9	8	1	5	6

✓ SOLVED